VINEET MADASSERI PAYYAPPALLI

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PROFILE

Energetic and innovative operations researcher/data scientist with strong mathematical modeling, optimization, and statistical data analysis skills. Seeking Operations Research Scientist/Data Scientist/Quantitative Analyst position, to apply analytics, data science, and quantitative modeling skills for developing smart solutions to challenging problems.

EDUCATION

University at Buffalo, The State University of New York *Aug 2013 – Jan 2019 (expected)*
Doctor of Philosophy (Ph.D.) in Industrial Engineering (Operations Research), pursuing GPA: 3.87/4.00
Master of Science (M.S.) in Industrial Engineering (Operations Research) GPA: 3.86/4.00
National Institute of Technology (NIT) Calicut, India *Jun 2005 – May 2009*
Bachelor of Technology (B.Tech) in Mechanical Engineering GPA: 7.09/10.00

RELEVANT SKILLS

Machine learning coursework: Statistical Data Mining I (supervised learning) and II (unsupervised learning)
Optimization & modeling: Operations research, Statistics, Predictive modeling, Risk analysis, Game theory
Programming & analytics tools: R, Matlab, JAVA, CPLEX, Minitab, SQL, PostgreSQL, Python, Jupyter, RShiny
Simulation & other tools: ARENA, AnyLogic, @Risk, Decision Tree, LaTeX, MS Excel
Personal: Analytical thinking & writing, Public speaking, Team work & leadership, Creating & keeping deadlines

EXPERIENCES & PROJECT

(1) Data Science Intern, WestRock*, Atlanta, Georgia* *May 2018 – August 2018*
Performance improvement of Supply Chain Optimization model [Tools used: CPLEX, Python, R, Jupyter] Developed applications in: (i) CPLEX Python API for batch-solving of sub-problems and prototype testing of distributed parallel computing; (ii) R for data analysis and visualization; and (iii) R and Python to identify "clusters" of machines that would help parallelization.

(2) University at Buffalo, Research/Teaching Assistant*, Buffalo, New York* *Aug 2014 – Present*
Warehouse and delivery improvement project at Meals on Wheels for Western New York [Tools used: MS Excel] (Pro-bono project, University at Buffalo) *Mar 2017 – Present*
Objectives: (i) to measure and improve warehouse capacity and utilization, by analyzing inventory data; and (ii) proposed a check-list mechanism that reduced incorrect/missing meal deliveries by 73.3%, during pilot test.

Modeling the cost effectiveness of fire protection resource allocation in the United States [Tools used: R, Matlab, Python, Jupyter, RShiny, LaTeX] (Research project, University at Buffalo) *May 2016 – Present*
Developed data-driven models to estimate the effectiveness of investment and to optimize resource allocation in fire protection, for mitigating fire losses. Currently working on spatio-temporal prediction of fire occurrence/fire risk. Data used: NFIRS (26 million incident records/year); firefighter data (30,000 fire departments); population, housing, economy, and other social/economic data (from U. S. Census Bureau).

Member of a finalist team in the Dow Big Data Challenge, organized by Dow Chemical Company and Texas A&M University [Tools used: R, MS Excel] (Competition, inter-university). *Mar – Apr 2016*
Worked on Dow's 18-month transactional orders data (structured data, with more than 2 million rows). Used ARIMA, Hidden Markov Model, and Exponential Smoothing in R to predict the likelihood for which customers are going to order which products in a given month; and forecast the quantity of orders.

Game-theoretic cyber information sharing model using particle swarm optimization algorithm [Tools used: Matlab, LaTeX] (Course project, University at Buffalo). *Jan 2016 – May 2016*
Created a mathematical model and algorithm using gradient-based and multi-objective particle swarm optimization concepts for simulating the decision making of governments and private companies in cyber threat info sharing.
Behavioral Analysis of Illegal Fishery in the Gulf of Mexico with Random Forest [Tools used: R] (Course project, University at Buffalo). *Aug 2015 – Dec 2015*
Compared the performance of various supervised learning models (linear and logistic regression, kNN, LDA, QDA, rpart, and random forest) in predicting illegal fishing in the Gulf of Mexico, using data from the U.S. Coast Guard.
Modeling deterrence and risk preferences in a sequential attacker-defender game with continuous defense effort [Tools used: Matlab, LaTeX] (M.S. Thesis, University at Buffalo). *Aug 2014 – May 2015*
Created a game-theoretical model that incorporated risk preferences of defenders/attackers. Identified specific scenarios in which this model performed better than conventional risk-neutral models.
Teaching assistant for graduate course: Applied Stochastic Processes (IE576) in Spring 2016 and Spring 2017; and for undergraduate courses: Applied Probability and Statistics Inference (EAS305) in Fall 2014, and Simulation Modeling & Analysis (IE477) in Fall 2017.

(2) Areva/Alstom T&D. Manufacturing & Quality Control Engineer, *Chennai, India* *Jul 2009 – Jun 2013*
Reducing assembled products inventory [Tools used: MS Excel] (Project, Alstom T&D). *Jan 2011 – Dec 2012*
Reduced assembled products inventory by 75%, which was about 11.438 million INR (0.18 million US Dollars) from an initial 15 million INR (0.24 million US Dollars), in 2 years.
Improving inventory turns ratio, and minimizing stock-outs [Tools used: Minitab, MS Excel] (Project, Areva T&D). *Jul 2009 – Jun 2010*
Proposed statistical safety stock calculations to improve inventory turns ratio about three-fold (from 3 to 10) and to reduce risk of stock-outs for critical components.

PUBLICATIONS

Title: **Modeling and Validating Public-Private Partnerships in Disaster Management – A Sequential Game Considering Risk Preferences.** *Journal:* Decision Analysis. *Authors:* Peiqiu Guan, Jing Zhang, Vineet M. Payyappalli, and Jun Zhuang. *In press, 2018.*
Title: **Total Cost of Fire in the United States**. *Technical Report*, National Fire Protection Association. *Authors:* Jun Zhuang, Vineet M. Payyappalli, Adam Behrendt, and Kathryn Lukasiewicz. *Published:* October 2017.
Title: **Deterrence and Risk Preferences in Sequential Attacker-Defender Games with Continuous Efforts**. *Journal:* Risk Analysis. *Authors:* Vineet M. Payyappalli, Jun Zhuang, and Victor Richmond R. Jose. *Vol:* 37. *No:* 11. *Pages:* 2229-2245. *Year:* 2017.

AWARD

Runner-up for Decision Analysis Society Best Student Paper Award *Oct 2017*
Selected by the Decision Analysis Society of the Institute for Operations Research and the Management Sciences (INFORMS) as a runner up (from among 20 applicants) for the best student paper in 2016.

PROFESSIONAL ACTIVITIES

President* of Student Chapter of INFORMS, University at Buffalo *Aug 2017 – May 2018*
Responsibilities: (1) Provide direction to the chapter's activities through leadership and team work. (2) Identify and develop student projects with local not-for-profit organizations. Organized a record 36 events, leading to INFORMS awarding the chapter Summa Cum Laude, the highest level of achievement for a student chapter, for 2017.
*Served as President-Elect (Sep 2016 - Aug 2017), and as Vice President (Aug 2014 - Sep 2016).